3 Simple Raspberry Pi Sense HAT Projects for Kids and Families

By Les Pounder published May 03, 2020

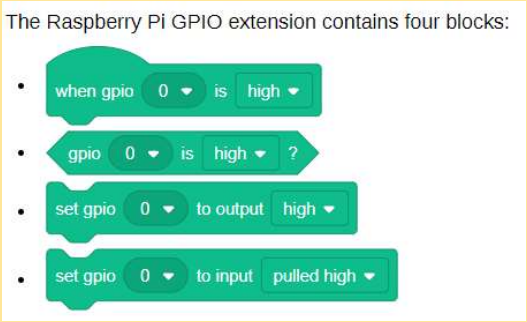
https://www.tomshardware.com/how-to/raspberry-pi-sense-hat-projects-for-kids

Prepared by: Dr. Ming-Shan Su

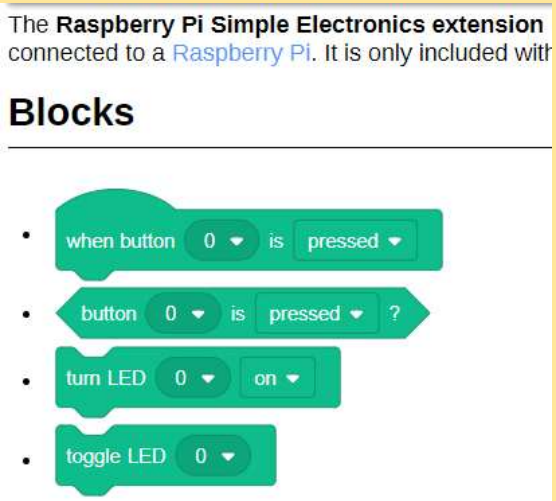
The Raspberry Pi Sense HAT extension contains 22 blocks:

Hands-on physical computing projects

GPIO:



Electronics

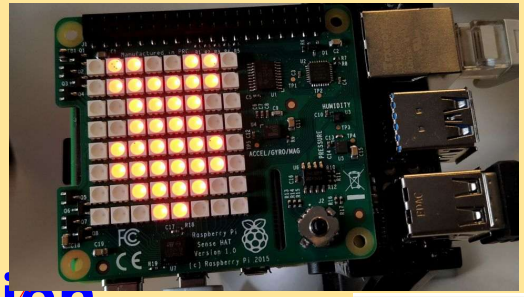


Pre-Configuration

Hands-on physical computing projects

Setting Up the Raspberry Pi Sense HAT and Scratch 3

1. **Attach the Sense HAT**
2. **Enable I2C in raspi-config (optinal)**
3. **Install Scratch 3**
4. **Install the Sense HAT Scratch 3 extension**



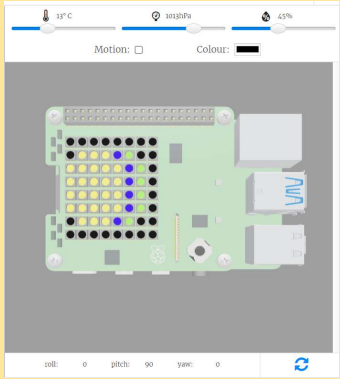
Note:

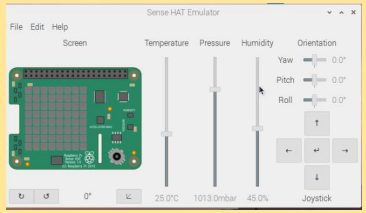
a. You can only try this project on the Raspberry Pi device

b. The emulator

   a. One on https://trinket.io/sense-hat works only with Python, not Scratch

   b. Another, Scratch first and add SenseHat, then Emulator will pop up.





https://youtu.be/rDA1xkpt7wo [2:51] https://youtu.be/B36czuDW-Sg [2:23, my]

P-01: Scrolling "Hello World" App with Raspberry Pi Sense HAT (step-0 warm-up)

0.a connect to the same wireless (or wired) networ

ping -4 pi0xx (e.g. pi002, see the label on the s

0.b write down the IP address (e.g., 192.168.0.101)

0.c use VNC viewer to have a monitor-less/Headles

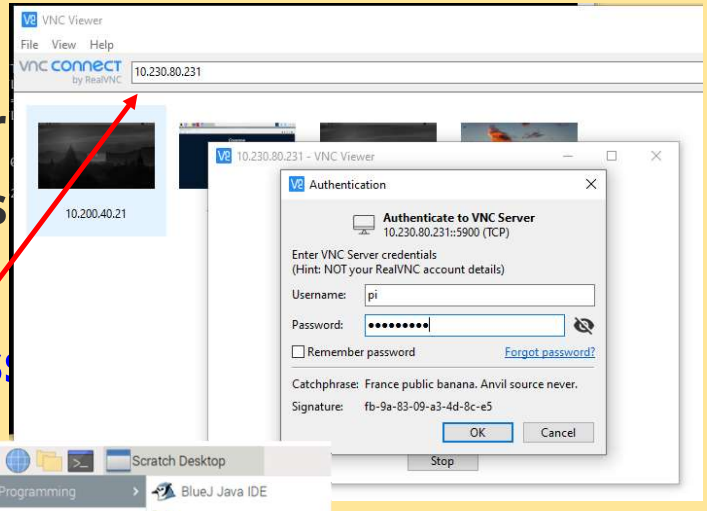
user-name: pi, password: raspberry

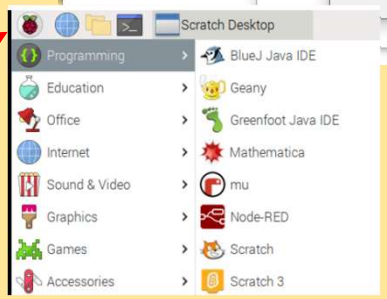
0.d click R-Pi-icon -> Programming -> Scratch 3

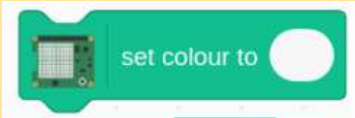
0.e run Scratch 3 -> add SenseHAT extension

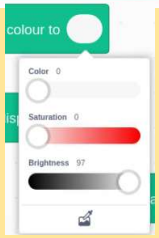
0.f Testing: Drag the "display text Hello!" f

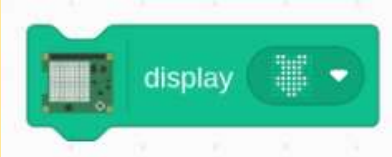
0.g: Cool!

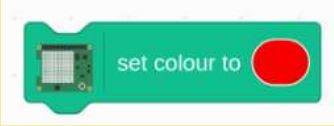






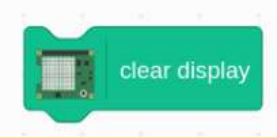


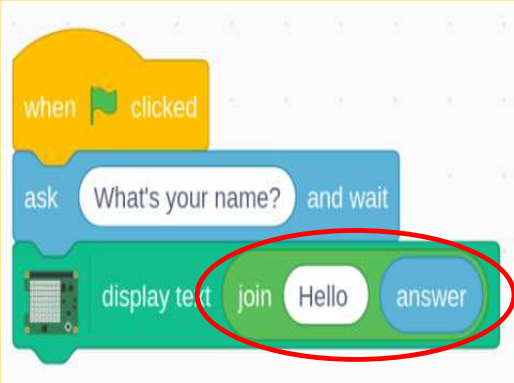




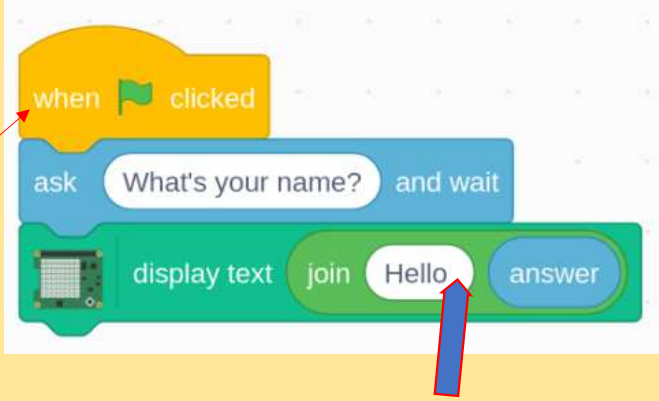
P-01: Scrolling "Hello World" App with Raspberry Pi Sense HAT

1. **Drag "When Green Flag is clicked"** from the Events section into the coding area.
2. **Drag the "What is your Name" block** from "Sensing" and connect it under the previous block.
3. **Connect the "Display Text" block** from the Sense HAT section to the "what is your name" block.
   1. Test
   2. We want to say hello to the user, but we also want to call them by their name.
4. **Find the "join apple banana" block** under Operators and drag it over to **replace the word "hello"** in display text.

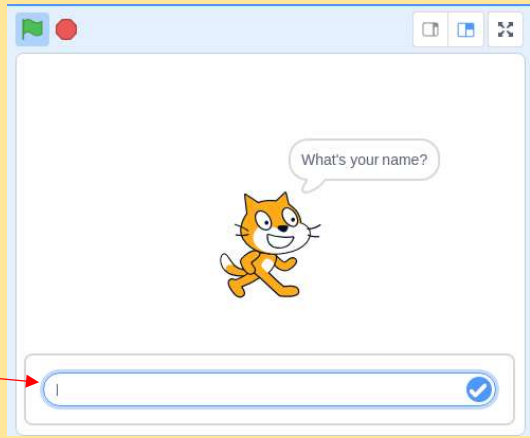




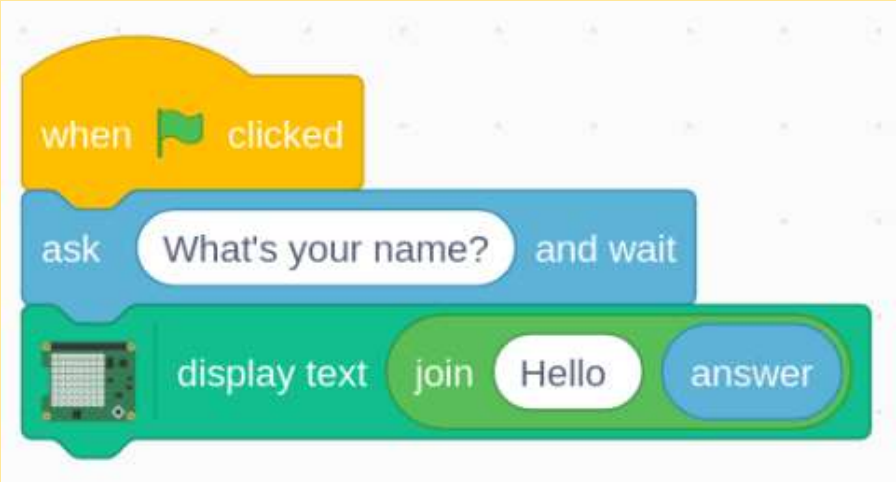
Scrolling “Hello World” App (

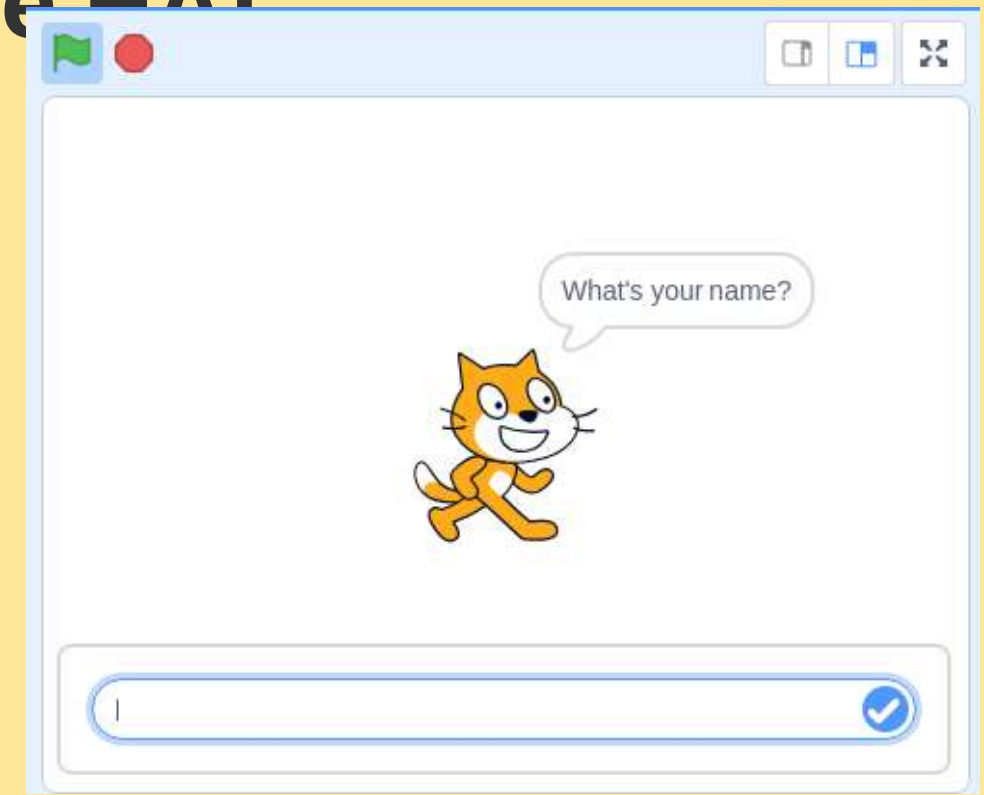


5. **Change “apple” to “Hello "** (the space after Hello is important!).
6. **Drag the “answer” block** from Sensing and **drop it on top of the “banana”** section of the join block.
   - By doing this the answer to the question “What’s your name?” is saved as a variable called “**answer**” which is then used to greet the user by their name.
7. **Click on the Green Flag icon**, just above the stage and the sprite will ask your name.
8. **Enter your name** and watch as your Sense HAT greets you by name.



Scrolling “Hello World” App with Raspberry Pi Sense HAT



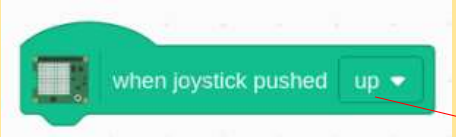


Done Project-01.
Fun and Interesting!

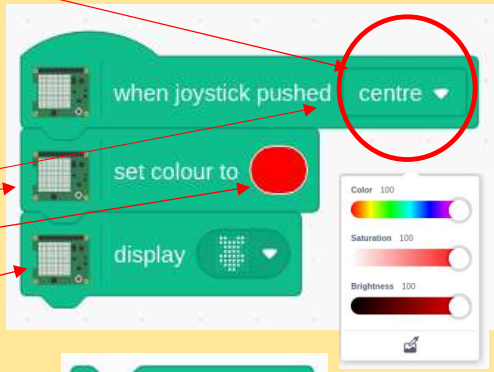
- Done Project-01.
- Fun and Interesting!

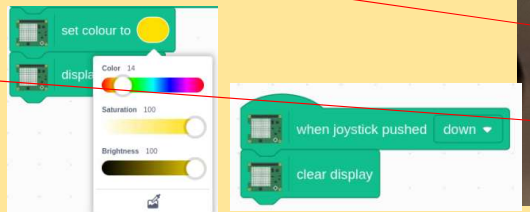
- Let’s Try More …
  - Temperature sensor

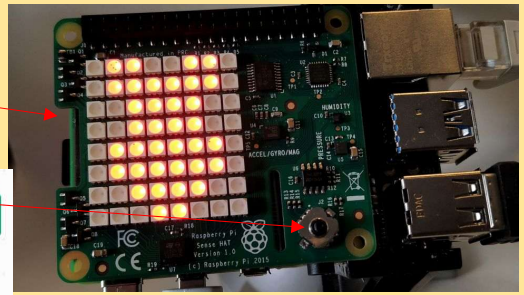
P-02: Joystick Input, Temperature Reading with Raspb nse HAT



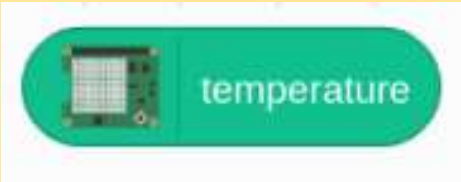
1. The test project uses the "When joystick pushed up" block found in the "Sense HAT" section.
2. **Drag the "When joystick pushed up" block** from the "Sense HAT" section into the coding area then **change the direction** to push the joystick from up to **centre**.
3. **Add the "set color to" block**, also from the "Sense HAT" section. Then **change the color** to something interesting (e.g., **Red**).
4. **Connect the "Display..." block** to the previous block. The default image is **some pixel art** of the **Pi logo**.
5. **test** by pressing the **joystick centre**



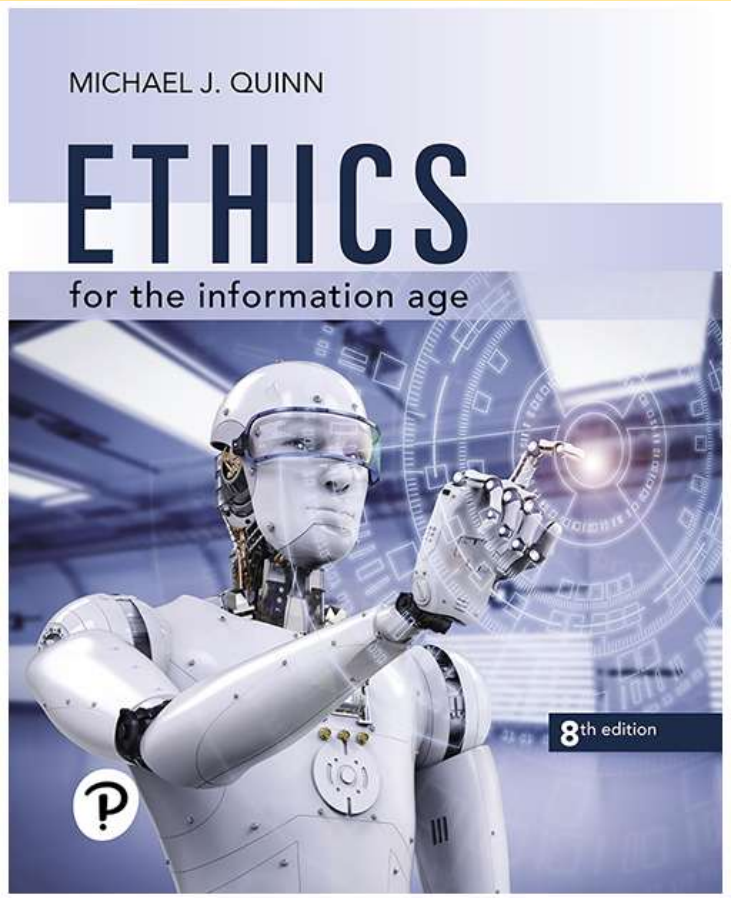




- A more **complex** project. This project will check the **temperature**
- **Click the temperature block (i.e., report-block) to have reading**
- **Why only 39.94?**
  - **Sensor Problem/Mal-function? Or**
  - **Locality/Region/Country Issue?**
- **Observation**:
- In U.K, the temperature unit is based on degree C (i.e., Celsius)
  - In addition, made in USA?, no, made in UK
- In U.S? degree F (Fahrenheit)
- **Other examples?**
  - **Pound vs. Kilogram**
  - **Inch vs. centimeter**
  - **Big Deal? Yes, it is.**



Side-Note:



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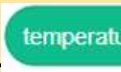
8.4.4 **Robot Missions to Mars**
NASA designed the $125 million Mars Climate Orbiter to facilitate communications between Earth and automated probes on the surface of Mars, including the Mars Polar Lander. Ironically, the spacecraft was lost because of a miscommunication between two support teams on Earth.

The Lockheed Martin flight operations team in Colorado designed its software to use **English units.** Its program output thrust in terms of **foot-pounds.** The navigation team at the Jet Propulsion Laboratory in California designed its software to use **metric units.** Its program expected thrust to be input in terms of newtons. One foot-pound equals 4.45 newtons. On September 23, 1999, the Mars Climate Orbiter neared the Red Planet.

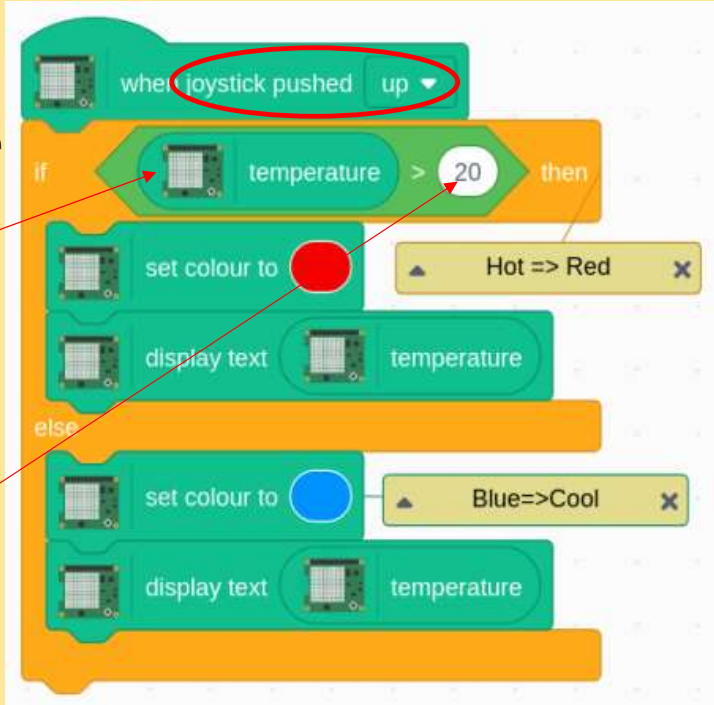
When it was time for the spacecraft to **fire its engine to enter orbit**, the Colorado team supplied thrust information to the California team, which relayed it to the spacecraft. Because of the units mismatch, the **navigation team specified 4.45 times too much thrust. The spacecraft flew too close to the surface of Mars and burned up in its atmosphere.**

A more complex project. This project will check the temperature

- Using the onboard temperature sensor of the Sense HAT. It will save the temperature to a ***variable***, then compare that variable to see if it is **greater** than 20 degrees Celsius. If so, then the **text scrolling across** the Sense HAT will be in **Red**. If below 20C then it will be **Blue**.



1. **Drag the "When joystick pushed up"** into the coding area.
2. **Add the "If, then, else" block** and connect it to the previous block.
3. **Drag "_ > 50"** and **place it inside the blank shape** between "if and then,"
4. **Drag the "temperature" variable block** and place it over the first blank of "_ > 50" and then change the 50 to **20**



A more complex project. This project will check the temperature

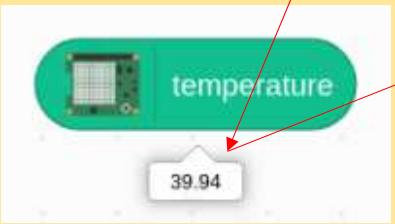
5. Complete the rest of the code.
6. Test and Debug, if any.

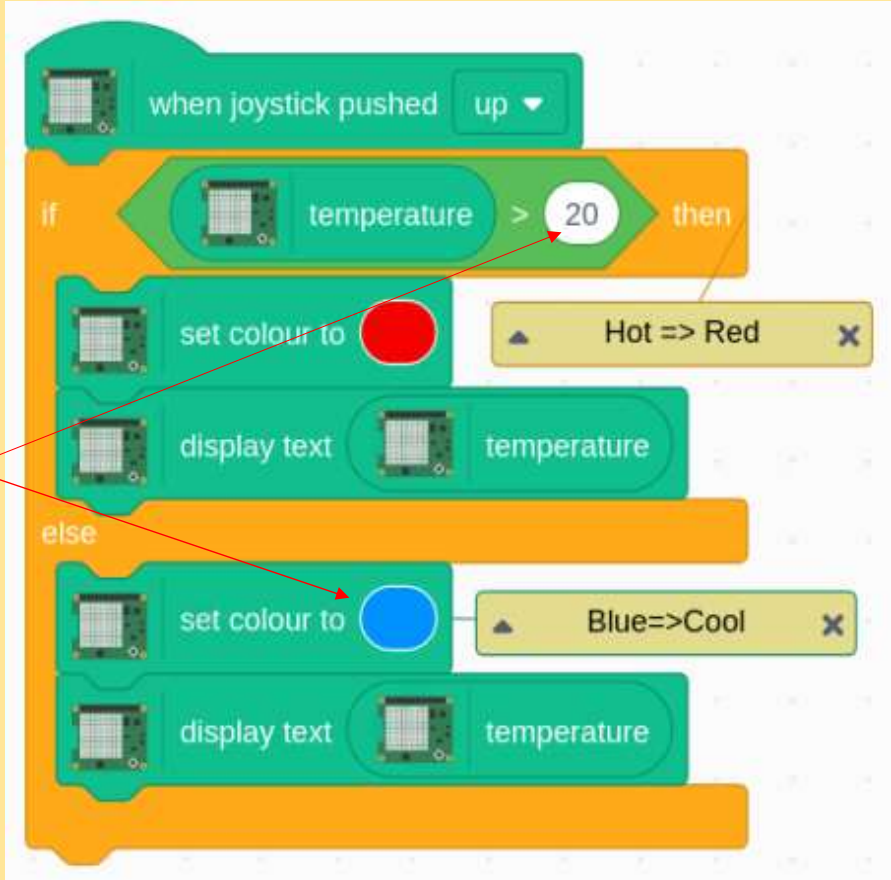
   * make sure the **blue-text** will be displayed too.

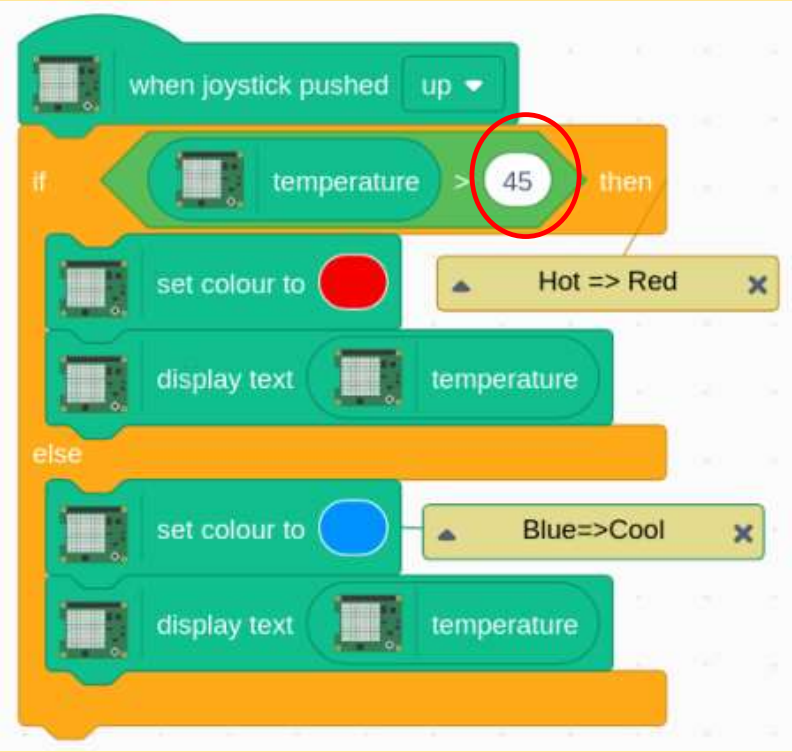
7. Something is not quite right!

Why?

We are in Durant, not UK, so

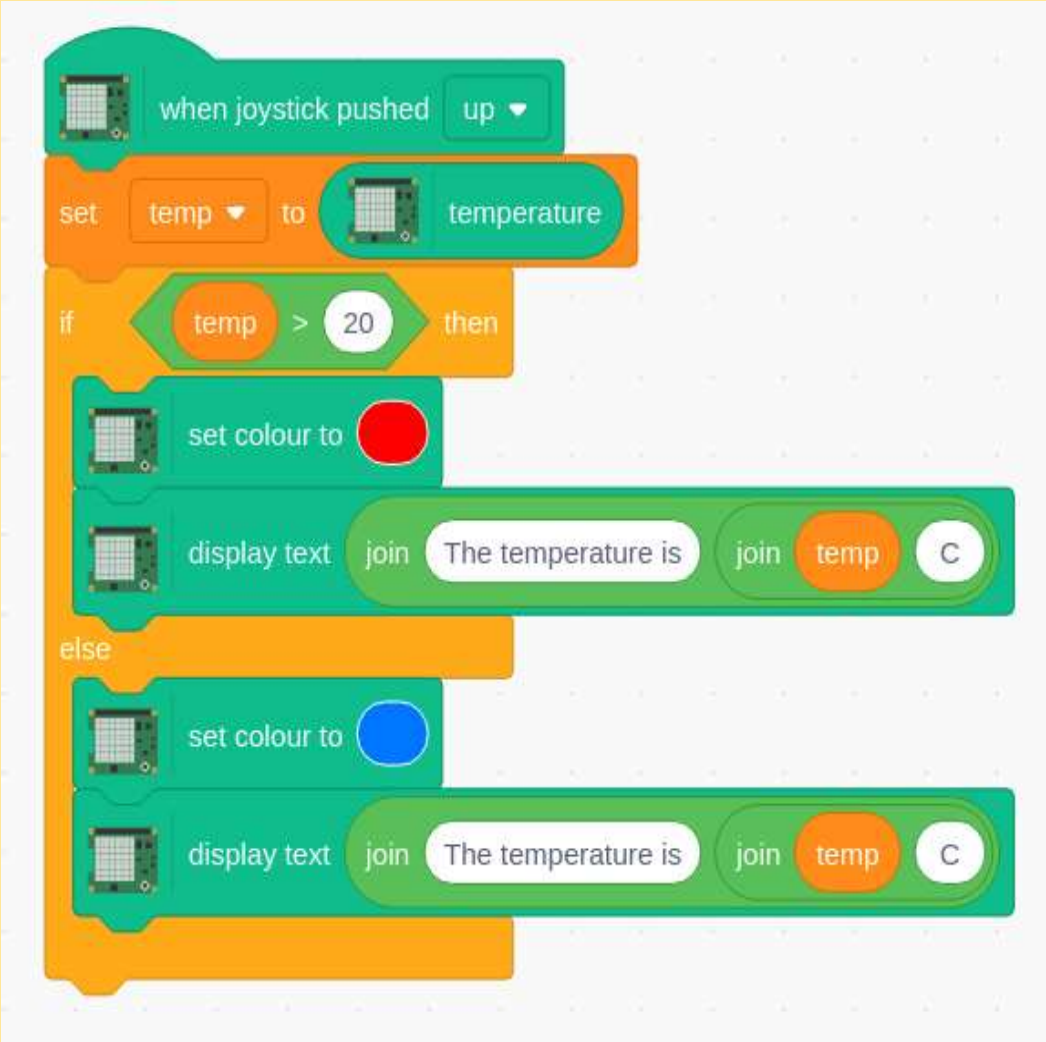




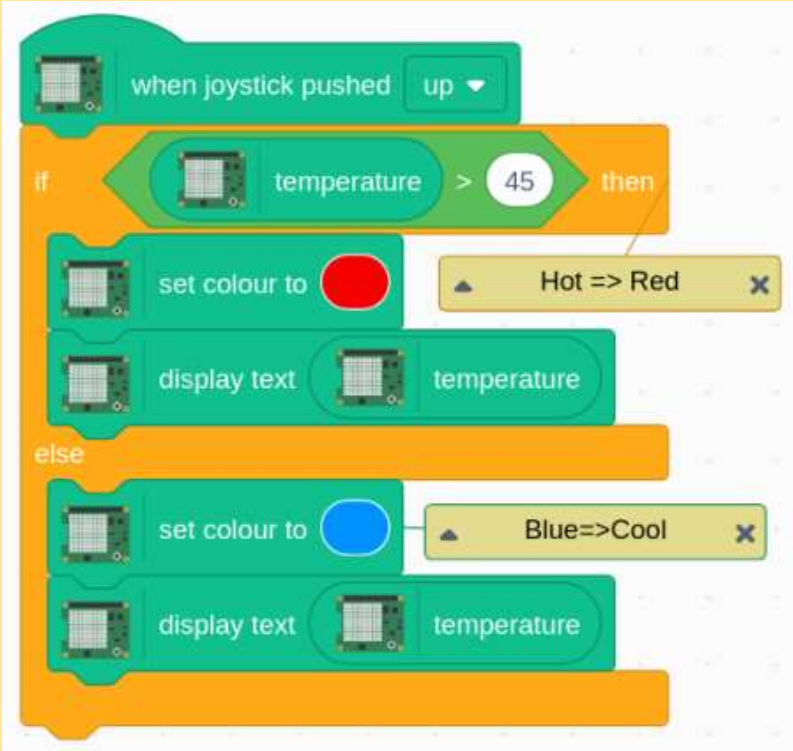


- Applications in life?
  - Alarm, Gauge (Red/Blue/Green/?) or
    - Colored-Coded
  - Fire?
    - Sound the Alarm or …
  - Notification (text, email, …)?

- Author’s original code.

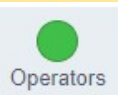


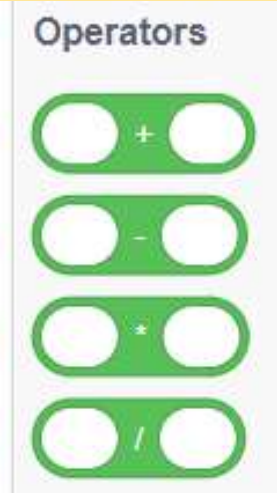
I like to keep it simple for beginners.



Challenge for you, not homework!
Conversion between temperature units (C vs. F)

- Formula: degree_C x 9/5 +32 => degree_F
  - (0°C × 9/5) + 32 = 32°F





https://www.google.com/search?q=celsius+to+fahrenheit&rlz=1C1GCEU_enUS861US861&oq=celsi&aqs=chrome.0.35i39j69i57j0i131i433i512l2j46i131i199i433i465i512j0i433i512l3j0i131i433l2.3319j0j15&sourceid=chrome&ie=UTF-8

- I just shared partial teaching materials. Please refer to [1] below to develop more Hands-on projects or activities to meet your needs.

Reference:

- [1] 3 Simple Raspberry Pi Sense HAT Projects for Kids and Families, By Les Pounder published May 03, 2020,
  - https://www.tomshardware.com/how-to/raspberry-pi-sense-hat-projects-for-kids